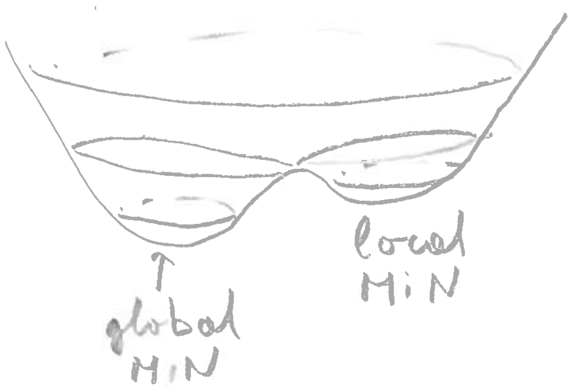
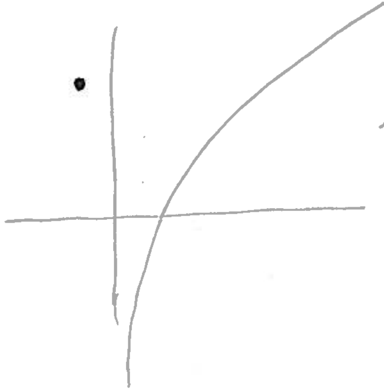
12.8 Max & Min

Defn Let f be a function defined on a region S and $P_0 \in S$.

- $f(P_0)$ is a <u>global max</u> of f on S if $f(P_0) \geq f(P)$ for all $P \in S$
- $f(P_0)$ is a <u>global min</u> if $f(P_0) \leq f(P)$ for all $P \in S$
- $f(P_0)$ is an <u>extreme value</u> if $f(P_0)$ is either a global max or global min
- $f(P_0)$ is a <u>local max/min</u> if there exists a neighborhood $P_0 \in N \subset S$ s.t. $f(P_0)$ is a global max/min on N.



Thm If f is continuous on a closed bounded set S, then f has a global max & min.



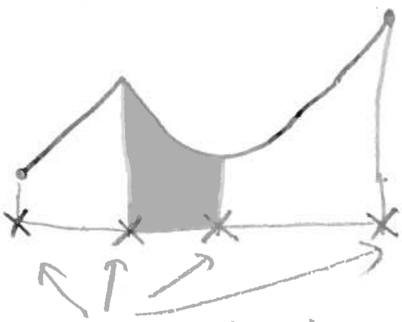
$\ln x$ no max/min on $(0, +\infty)$

$S = [1, 2]$ $\quad \max_{P \in S} \ln x = \ln 2$

$\min_{P \in S} \ln x = \ln 1 = 0$

Thm $P_0 \in S$. f defined on S. If $f(P_0)$ is an extreme value then P_0 is a critical point

critical points
- $\partial S = \{$ boundary pts of $S \}$
- Stationary points = pts in interior of S such that $\nabla f = 0$. Tg plane is horizontal
- Singular pt = pts in interior of S s.t. f is not differentiable



Critical pts.

Example: Find max & min of $f(x,y) = x^2 - 2x + y^2$ on $x^2 + y^2 \leq 100$.

Critical pts are $x^2 + y^2 = 100$ & $\nabla f = (2x-2, 2y) = (0,0)$

$\Rightarrow x = 1,\ y = 0$

$\Rightarrow f(1,0) = -1$

$x = 10\cos\theta$
$y = 10\sin\theta$

$x^2 - 2x + y^2 = 100\cos^2\theta - 20\cos\theta + 100\sin^2\theta = 100 - 20\cos\theta$

$\frac{d}{d\theta}(100 - 20\cos\theta) = 20\sin\theta = 0 \Rightarrow \theta = 0, \pi$

$\theta = 0 \to (10,0)$ $\quad f(10,0) = 80$

$\theta = \pi \to (-10,0)$ $\quad f(-10,0) = 120$

so MAX $= 120$ at $(-10,0)$

MIN $= -1$ at $(0,0)$

If $S = \mathbb{R}^2$ then MIN $= -1$, no MAX

$x^2 - 2x + y^2$ goes to $+\infty$ as $(x^2+y^2) \to +\infty$

Ex Find critical pts, max, min for $f(x,y)=x^2-y^2$

$\nabla f = (2x, -2y)$ $\qquad$ $(0,0)$ only critical pt

$f(0,0)=0$

$x=0 \quad f(0,y) = -y^2 \longrightarrow -\infty$ as $y \to +\infty$

$y=0 \quad f(x,0) = x^2 \longrightarrow +\infty$ as $x \to +\infty$

NO MAX, NO MIN.

If $S=\{x^2+y^2 \le 1\}$ then $\partial S = \{x^2+y^2=1\}$ are critical pts.

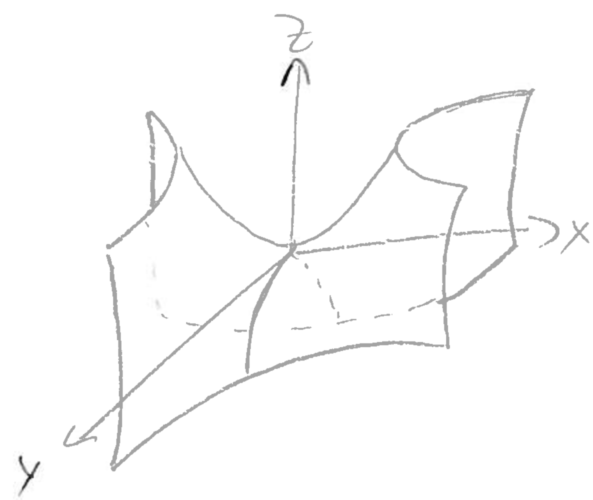
$x=\cos\theta,\ y=\sin\theta$

$f(x,y) = \cos^2\theta - \sin^2\theta = 1-2\sin^2\theta$ $\qquad$ $\frac{df}{d\theta} = 4\sin\theta\cos\theta$

$\frac{df}{d\theta}=0 \qquad \theta = 0, \pi, \pi/2, \frac{3\pi}{2}$

$x,y = (1,0)\ (-1,0)\ (0,1)\ (0,-1)$

$f(x,y) = \underbrace{1 \quad 1}_{MAX} \quad \underbrace{-1 \quad -1}_{MIN}$



Second partial test: Suppose that $f(x,y)$ has continuous 2nd order partial derivatives $f_{xx}, f_{xy}, f_{yx}, f_{yy}$ near (x_0, y_0) and $\nabla f(x_0,y_0) = \vec{0}$. Let $D = f_{xx}(x_0,y_0) f_{yy}(x_0,y_0) - f_{xy}^2(x_0,y_0)$

Then
- If $D>0$ & $f_{xx}(x_0,y_0) < 0$ we have local MAX
- If $D>0$ & $f_{xx}(x_0,y_0) > 0$ " " " MIN
- If $D<0$ we have a saddle
- If $D=0$ the test is inconclusive

Eg $z = -x^2 - y^2$ $\nabla f = (-2x, -2y)$ $\nabla f(0,0) = (0,0)$

$D = ((-2)\cdot(-2) - 0^2) = 4 > 0$ $f_{xx} = -2 < 0$ Local MAX

$z = x^2 + y^2$  $\nabla f = (2x, 2y)$ $\nabla f(0,0) = (0,0)$

$D = (2\cdot 2 - 0^2) = 4 > 0$ $f_{xx} = 2 > 0$ Local MIN

$z = x^2 - y^2$  $\nabla f = (2x, -2y)$ $\nabla f(0,0) = (0,0)$

$D = (2\cdot(-2) - 0^2) = -4 < 0$ Saddle

Ex Find extrema of $3x^3 + y^2 - 9x + 4y$

$\nabla f = (9x^2 - 9, 2y + 4)$ $\nabla f = 0$ $x = \pm 1$, $y = -2$

$D = (18x \cdot 2 - 0^2) = 36x$

$D(1,-2) = 36$ $f_{xx}(1,-2) = 18$ Local MIN $f(1,-2) = 3 + 4 - 9 - 8 = -10$

$D(-1,-2) = -36 < 0$ Saddle

$y = 0$ $3x^3 - 9x$ $\to +\infty$ if $x \to +\infty$, $\to -\infty$ if $x \to -\infty$ No global MAX/MIN

Ex 1 Find critical pts & indicate if they are max, min saddle or undetermined.

a) $z = x^2 + 4y^2 - 4x$ b) $z = 2x^4 - x^2 + 3y^2$ c) $z = xy$

d) $z = xy + \frac{2}{x} + \frac{4}{y}$ e) $f(x,y) = \cos x + \cos y + \cos(x+y)$

Ex 2 Find global max & min.

a) $f(x,y) = 3x + 4y$ $S = \{(x,y) \mid 0 \le x \le 1, -1 \le y \le 1\}$

b) $f(x,y) = x^2 + y^2$ $S = \{(x,y) \mid -1 \le x \le 3, -1 \le y \le 4\}$

c) $f(x,y) = x^2 - y^2 + 1$ $S = \{(x,y) \mid x^2 + y^2 \le 1\}$

Ex 3 Find the shortest distance from the origin to the plane $x + 2y + 3z = 12$

Ex 4 Find the minimum distance from $(1,2,0)$ to $z = x^2 + y^2$